

RICD Collapse-Causal Tracker (C-CT): What It Is

A description document — distinct from the Findings, Narrative, and Process Log reports

This document explains what a collapse-causal tracker is as a category of thing: what it measures, what RICD math it draws on, what it is for, and how the one built so far (the higher-education tracker) generalizes to the domains planned next. It is not a repeat of the findings, the narrative of how those findings were reached, or the process log; those already exist and are not remade here.

What a Collapse-Causal Tracker Measures

A collapse-causal tracker works at one specific level in a three-layer model of how a collapse actually happens:

1. Mechanism: the actual causal actions of an identifiable agent, human or non-human, that set a collapse in motion.
2. Second order effect, functioning as general cause: the aggregate, measurable consequence those actions leave behind in a system's own trajectory.
3. Flip: the classified outcome itself: collapsed, or not.

A collapse-causal tracker operates entirely at layer 2 and classifies layer 3 from it. It measures the second-order effects a cause leaves behind, not the mechanism's first-order action itself, but the compounding, downstream trace that action leaves in a system's resource trajectory, its informational divergence, its latent stress and resource debt, and it does this without ever needing to know, or claim to know, what the underlying mechanism was, or whether a human decision or a non-human process drove it. That is a deliberate scope limit, not an oversight: identifying the specific agent responsible is left to a different kind of tracker (see below).

The RICD Math It Uses

The tracker implements a real subset of the RICD manuscript (specifically Parts 6, 7, 8, and 10.5b) against real institutional data. The manuscript itself is domain-independent and covers considerably more than this tracker uses.

Feedback Divergence Field Mechanics (FDFM): three representations, and the gap between them

Every tracked system carries three parallel readings of its own state: an official representation (O_o , what is claimed or reported), an operational representation (O_p , an honest, instantaneous projection of the true underlying state), and a meta-representation (P_m). The divergence potential, $\Phi = D_{op} + D_{om} + D_{pm}$, sums the Jensen-Shannon divergence between each pair, and measures how far these three readings have pulled apart from one another.

The official channel is deliberately modeled as its own independently-evolving, lagged tracking of operational reality — $O_o(t+1) = O_o(t) + \kappa(O_p(t) - O_o(t))$ — rather than a second, static rescoring of the true state. That is what “official messaging lags or suppresses reality” actually

means mechanically, with κ as a controllable knob running from healthy (official tracks operational reality closely) to shielded (official is suppressed from tracking real change). This is also what produces the false-stability signature the tracker is built to catch.

Resource dynamics

Resources update through a maintenance-first, clipped rule: an institution's maintenance obligations are met first out of whatever resources are available, and only what remains funds adaptive spending. This models where the slack actually runs out under resource stress, rather than assuming all spending is equally discretionary.

Permeability and regime classification

A mean-reverting boundary-permeability parameter (π_f) governs how exposed the system is to shocks passing through it. Regime classification compares the causal — trailing-only, differenced — variance of the official channel against the operational channel: when the operational channel is far more volatile than the official one, the period is classified high-entropy (official-shielded); the reverse is medium-entropy; comparable variances are baseline.

Collapse onset

A hull-breach condition flags when divergence exceeds what current permeability can absorb, but onset is only confirmed once a breach persists for a minimum number of consecutive periods (a single spike does not register as onset).

All of this instantiates the identity-metric simplification (v0.1) rather than the manuscript's full, general metric machinery; a stated, flagged simplification, not a silent one, and one of the things a future adapter could choose to revisit rather than inherit unexamined.

Distinguishing C-CT from C-CAT

C-CT and the collapse-causal agent tracker (C-CAT, described in its own seed note) are adjacent layers of one stack, not competing approaches to the same problem. C-CT works entirely on layer 2 (effect / general cause) data and is agnostic to what produced it; it can flag that a system's trajectory looks collapse-like without ever knowing whether a person or a process caused it. C-CAT works one layer down, at layer 1 (mechanism), and asks what specific agent — human or non-human — actually produced the divergence C-CT measures. C-CT is built to run continuously across a whole panel from routinely-reported data; C-CAT is built to be pointed at a specific, already-flagged case and dig into what produced it.

What Got Fit for the Higher-Ed Panel

The one collapse-causal tracker built so far applies this framework to U.S. higher education. Eight features, extracted from real IPEDS and finance data, were fit against a validated, 54-institution panel (23 confirmed closures spanning seven distinct collapse mechanisms, 31 confirmed-stable comparisons):

- `d_A_trend`, `d_A_final` — the trend and current level of an institution's informational-divergence trajectory.
- `delta_R_final`, `delta_R_trend` — the severity of resource divergence at the most recent point, and its own trend.
- `debt_spike` — the debt-shock signature.
- `frac_high_entropy` — the fraction of recent periods classified high-entropy by the regime test. As of a fix applied 2026-09-19, this is gated to zero whenever `debt_spike` is not positive, because the underlying variance measure cannot on its own distinguish a large positive shock from a large negative one; the gate gets the panel back to 100% leave-one-out accuracy, but it is a stopgap on the sign-blindness problem, not a full, direction-aware reconstruction of the feature.
- `reserve_adequacy` — $\log(1 + \text{endowment} / \text{peak enrollment})$, scaled by enrollment rather than by debt specifically so it holds across institutions of very different absolute size.
- `research_ratio` — research expenditure relative to instructional expenditure.

Alongside these eight fitted features, a small number of institutions with a real, independently confirmed governance verdict (an accreditor's show-cause order or accreditation withdrawal) are classified by direct override rather than by the fitted statistical model at all, a deliberate design choice, confirmed to work better than fitting governance as a weighted feature, not an oversight. It is worth being explicit about this when citing the panel's 100% leave-one-out accuracy: those override cases were never actually put to the statistical classifier, so the figure describes the combined system — eight fitted features plus a governance bypass — rather than the predictive power of the eight features alone.

How This Generalizes to Other Trackers

The practical engineering discipline behind building this tracker is written up separately, in the RICD Adapter Instructional Manual, and is not reproduced here; however, it's worth summarizing here, as it's what the next adapter needs to inherit:

- Verify what a data field actually measures for its specific reporting year (codes and forms change meaning across years and institution types without warning).
- Classify each raw observable as drift-type or shock-type before choosing a modeling tool for it; forcing a shock-type quantity through a drift-type tool systematically understates its real severity.
- Validate every candidate feature, and every label, against real held-out cases; a theoretically well-motivated feature can still make overall performance worse, and a persistently misclassified case is as likely to be a labeling error as a model failure.
- Normalize by a genuinely independent scale measure, never by a quantity that itself moves with whatever is being normalized.
- Prefer a real, documented relationship over an invented one when modeling coupling between two systems.

The same three-part shape used here — a domain-specific adapter mapping real data into RICD's representations, a small fitted feature set validated against real, independently confirmed outcomes, and a governance-override-style bypass wherever a genuinely independent verdict exists for that domain — is the template for the domains currently planned next for RICD's collapse-causal tracking: banking, hospitals, public schools, and long-term care nursing homes. Each will need its own version of the eight-feature-equivalent fitting process and its own governance analogue, since what counts as an independent attesting body differs by domain, not a copy of the higher-ed panel's specific numbers or feature set.